MILITARY SPECIFICATION

RANGEFINDER, FIRE CONTROL: AN/VVG-1( )
11738801

This specification is approved for use by all Departments and Agencies of the Department of Defense.

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE EXERCISED IN OPERATING AND TESTING THE RANGEFINDER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, AND PAINT. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RANGEFINDER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers one type of rangefinder designated as Rangefinder, Fire Control: AN/VVG-1( ) Part Number (P/N) 11738801 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical

Military

MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of

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MIL-C-14880	Component Assembly: 10559479 (Transmitter)
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedures for
MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-C-50734	Control, Commanders, Laser Ranging: C-9134( )/VVG-1
MIL-B-50737	Power Supply, Battery: PP-6607( )/VVG-1
MIL-C-50742	Control, Receiver-Transmitter: C-8728( )/VVG-1
MIL-C-50743	Control, Power Supply, Laser: PP-9135( )/VVG-1
MIL-R-50746	Receiver-Transmitter, Laser: RT-1021( )/VVG-1
MIL-B-60621	Bonding Optical Assemblies (Glass to Metal)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

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MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

F8565556	Pressure Tester
F11738801	Rangefinder, Fire Control: AN/VVG-1

PACKAGING DATA SHEET

11738801	Packaging of Rangefinder, Fire Control: AN/VVG-1( )11738801

OTHER PUBLICATIONS

A11745424	Bonding of Optical Assemblies, Process for

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Fire Control Rangefinder AN/VVG-1( ), herein referred to as the rangefinder, shall be manufactured in accordance with Drawing F11738801 and drawings pertaining thereto. The components and cables listed in table I shall each have met the requirements of the listed specifications and drawings prior to assembly as the rangefinder. Except as specified herein, environmental tests are excluded at the component, assembly, and unit levels.

TABLE I
RANGEFINDER COMPONENTS AND CABLES

Item	Nomenclature	Part No.	Spec. No.
1	Receiver-Transmitter, Laser: RT-1021( )/VVG-1	11738821	MIL-R-50746
2	Power Supply, Battery: PP-6607/VVG-1	11738841	MIL-B-50737
3	Control, Commander's, Laser Ranging: C-9134 ( )/VVG-1	11738811	MIL-C-50734
4	Control, Power Supply, Laser: C-9135 ( )/VVG-1	11738831	MIL-C-50743
5	Control, Receiver-Transmitter: C-8728( )/VVG-1	11738851	MIL-C-50742
6	Cable, Interconnect - W50:	11738865	-
7	Cable, Interconnect - W51:	11738866	-
8	Cable, Interconnect - W52:	11738867	-
9	Cable, Interconnect - W53:	11738868	-
10	Cable, Interconnect - W54:	11738869	-
11	Headrest Assembly	11737449	-

3.2 General Specification. The rangefinder shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes

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d. Electrical and electronic assemblies

e. Part identification and marking

f. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844 except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.2.2 Optical bonding. Optical bonding shall be in accordance with MIL-B-60621 or document A1174542-4.

3.3 Performance. Unless otherwise specified, the rangefinder shall meet the performance requirements of this specification, except 3.3.1.6f (2), at standard ambient temperatures between +60 degrees Fahrenheit (^{o}F) and +90°F.

3.3.1 Power and signal. The rangefinder shall perform as specified herein when the power and signal of table II are applied as specified.

TABLE II
POWER AND SIGNAL

Item	Condition	Characteristics	Connections
1	Power Source:		Connected between the following terminals:
1.1	24 Vdc	Tolerance: ±6.0 Vdc Ripple: ≤200 m Vp-p	W50P1-52B and C (+) and W50P1-52D and E (-)
2.	Signal Source:		
2.1	Remote Range	+5.0 ±0.4 Vdc in series with 10 kilohm ±5 percent	Applied between A77J5-c and A77J5-e when specified herein.

3.3.1.1 Power turn-on condition. When the LASER MODE CONTROL switch on the commanders control is set to ON position, the following conditions shall exist:

a. The RANGE and RETURNS readout on the commanders control shall read zero and the LAST button shall be illuminated.

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b. The RESET button on the receiver-transmitter control shall be continuously illuminated.

c. The ON indicator light on the commanders control illuminate.

d. The RANGE button on the receiver-transmitter control shall begin to flash within 4 seconds.

3.3.1.2 Pulse forming network (PFN) charge and flashtube firing. Pressing the RANGE button when it is flashing (ranging) shall cause the PFN to discharge through the flashtube. The flashing red light on the RANGE button shall turn off during ranging and turn on steady after ranging, indicating that the PFN is recharged. Pressing the RANGE button after ranging when the red light is steady shall not affect the flashtube. When the PFN is charged, pressing the RESET button shall cause the red light on the RANGE button to flash and the range return selector LAST indicator lamp shall illuminate.

3.3.1.3 Target selection. Pressing the "1" or "2" RANGE RETURN SELECTOR button shall cause the range of the first or second target, respectively, to be displayed on the RANGE readout. When there are one, two, or three target returns, pressing the LAST RANGE RETURN SELECTOR button shall cause the range to the last target to be displayed. When there are four or more target returns, pressing the LAST RANGE RETURN SELECTOR button shall cause the range to the third target to be displayed.

3.3.1.4 RESET switch. Whenever the RESET button is pressed, the range RETURNS readout shall be zero, and the four RANGE readouts shall be zero.

3.3.1.5 LASER MODE CONTROL switch

3.3.1.5.1 TEST position. When the LASER MODE CONTROL switch is set to the TEST position, the RANGE button shall turn on steady and pressing the RANGE button shall not discharge the PFN through the flashtube. Also the requirements of 3.3.1.3 shall apply.

3.3.1.5.2 ON position. When the LASER MODE CONTROL switch is in the ON position and ranging is performed, the approximate range to target shall be read on the RANGE readout.

3.3.1.6 Ranging - laser firing

a. The rangefinder shall be capable of measuring target ranges to a range of 215 meters within the accuracy specified below.

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b. The rangefinder shall be capable of measuring ranges to a range of at least 4000 meters within the accuracy specified below. This may be simulated by an extinction test by placing neutral density filters over the PMT aperture with an attenuation of 36 dB.

c. The range accuracy shall be plus or minus 10 meters.

d. The range resolution shall be 20 meters.

e. Minimum time between ranging: When the RANGE button is depressed, the flashing red light on the RANGE button shall turn off and then turn on steady within 4 seconds.

f. Number of rangings on system internal power (battery fully charged): (1) 30 rangings minimum at +75 ±15°F at a rate of 3 per minute for maximum operating time of 10 minutes; (2) 10 rangings minimum at -20 ±2°F at a rate of 3 per minute for maximum operating time of 4 minutes.

g. The maximum number of rangings* shall not exceed 6 per minute for 2 minutes without deterioration of range accuracy with 3 minute intervals between each 2 minute ranging period.

h. The maximum sustained ranging* shall be 3 per minute over a period of one hour 15 minutes without deterioration of range accuracy.

3.3.1.7 Remote ranging. With the signal source of table II applied to the input pin as specified and ranging performed, the approximate range to target shall appear on the RANGE readout.

3.3.1.8 Peak output power. The peak output power (see 6.3.1) of the receiver-transmitter shall be not less than one megawatt.

3.3.1.9 Input voltage variation. The rangefinder shall meet the ranging accuracy specified in 3.3.1.6.c when the input voltage of the prime power source varies between +18 Vdc and +30 Vdc.

3.3.1.10 Maximum power drain. The maximum direct current drain from the prime power source shall be 12 amperes.

*Constant firing for 1.25 hours may be performed using a 30 minute rest period between each 1 hour of constant firing. The total lasing count per 24 hour time period shall not exceed 840.

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3.3.1.11 Primary power transient. The RANGE readout shall not change under the following conditions:

a. When the input voltage is dropped to $+6\ ^{+1}_{-0}$ volts for a duration of 100 ±10 milliseconds.

b. When the input voltage is increased to $+90\ ^{+0}_{-5}$ volts for a duration of 100 ±10 milliseconds.

3.3.1.12 Malfunction indication.

3.3.1.12.1 Overvoltage. When an overvoltage condition occurs in the PFN circuitry, the number "4" shall be displayed in the least significant digit position of the RANGE readout and the MALF lamp shall illuminate.

3.3.1.12.2 "A" trigger reply. When a malfunction occurs in the "A" trigger reply circuitry, the number "3" shall be displayed in the least significant digit position of the RANGE readout and the MALF lamp shall illuminate.

3.3.1.12.3 "A" trigger and counter. When a malfunction occurs in the reply gating or counter circuit card, the number "2" shall be displayed in the least significant digit position of the RANGE readout and the MALF lamp shall illuminate.

3.3.1.12.4 Battery charging. When a malfunction occurs in the battery charging circuitry, the number "1" shall be displayed in the least significant digit position of the RANGE readout and the MALF lamp shall illuminate.

3.3.1.13 Dimming (DMR) switch. Depressing the DMR switch on the commanders control shall cause all indicator lamps to change either from bright to dim or from dim to bright.

3.3.1.14 Test (TSW) switch. Depressing the TSW switch on the commanders control shall cause the number "8" to appear on the RANGE and RETURNS readouts and all indicator lamps shall light.

3.3.1.15 Sight/transmitter alignment. The transmitter beam shall be parallel to the receiver line of sight within 0.25 mil. The receiver line of sight shall be defined as the center of the 1 - mil reticle circle.

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3.3.1.16 Optical scan angles. The optical scan angles shall be as follows:

a. Elevation: Plus 20 degrees minimum in elevation and minus 10 degrees minimum in depression from a horizontal reference line. Horizontal reference shall be established by indication of the zero setting, plus or minus one degree, on the elevation control knob scale.

b. Azimuth: 20 degrees minimum total excursion plus and minus 10 degrees from a vertical reference line.

3.3.1.17 Image tilt. The image of a vertical line shall be vertical within 2 degrees throughout the elevation excursion range specified in 3.3.1.16a.

3.3.1.18 Image wander. The target image observed at the eyepiece shall not shift by more than ±2 degrees throughout the elevation excursion range specified in 3.3.1.16a.

3.3.1.19 Reticle illumination. With the reticle lamp energized, the reticle pattern shall be clearly defined in a darkened area. Rotation of the reticle light control knob from one extreme to the other shall increase or decrease the intensity of the reticle illumination.

3.3.1.20 Battery drain indicator. When the input voltage drops below +18 volts, the battery drain lamp shall illuminate, indicating system is functioning on secondary power.

3.3.1.21 Battery low indicator. When the internal battery is in the discharged state, the battery low lamp shall illuminate.

3.3.1.22 Drive train torque and backlash.

3.3.1.22.1 Azimuth. The aximuth drive train starting torque shall not exceed 50 ounce-inch (oz-in), when alternately applying torque in the positive and negative directions to the azimuth positioning knob of the receiver-transmitter control, and when monitoring for movement of the scan prism. Backlash (i.e., the sum of the positive and negative rotation) during the starting torque, excluding detent position, shall not exceed 40 degrees rotation of the azimuth positioning knob.

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3.3.1.22.2 Elevation. The elevation drive train starting torque shall not exceed 50 oz-in when alternately applying torque in the positive and negative directions to the elevation positioning knob of the receiver-transmitter control, and when monitoring for movement of the scan prism. Backlash during the starting torque shall not exceed 36 degrees rotation of the elevation positioning knob.

3.3.1.23 PFN discharge time. With the 24 Vdc power source (item 1.1 of table II) removed, the PFN current between pins t(+) and e(-) of test access connector J5 on the Laser Power Supply Control C-9135 ( )/VVG-1 (11738831) shall decrease to less than 7 microamperes within 25 seconds.

3.3.2 Environmental.

3.3.2.1 Storage temperature. The rangefinder shall meet the requirements of 3.3.1.3, 3.3.1.6b, 3.3.1.8, and 3.3.1.15 through 3.3.1.23 at ambient temperature after exposure to and thermal stabilization at -80° F and +160° F.

3.3.2.2 Operating temperature. The rangefinder shall meet the requirements of 3.3.1.3, 3.3.1.6c, and 3.3.1.8 while exposed to and thermally stabilized at -40° F and +125° F, subsequent to which the requirements of 3.3.1.3, 3.3.1.6b, 3.3.1.8, and 3.3.1.15 through 3.3.1.23 shall be met at ambient temperature. The rangefinder shall meet the requirements of 3.3.1.6f(2), while exposed to and thermally stabilized at -20° F.

3.3.2.3 Vibration "A". The rangefinder shall meet the requirements of 3.3.1.3 and shall show no physical failure during the following vibratory exposure:

a. Double amplitude and acceleration as shown in figure 1.

b. Total test time of 150 minutes per axis, except there shall be no resonance dwells.

Subsequent to vibration exposure, the rangefinder shall meet the requirements of 3.3.1.6b, 3.3.1.8, and 3.3.1.15 through 3.3.1.23 at ambient temperature.

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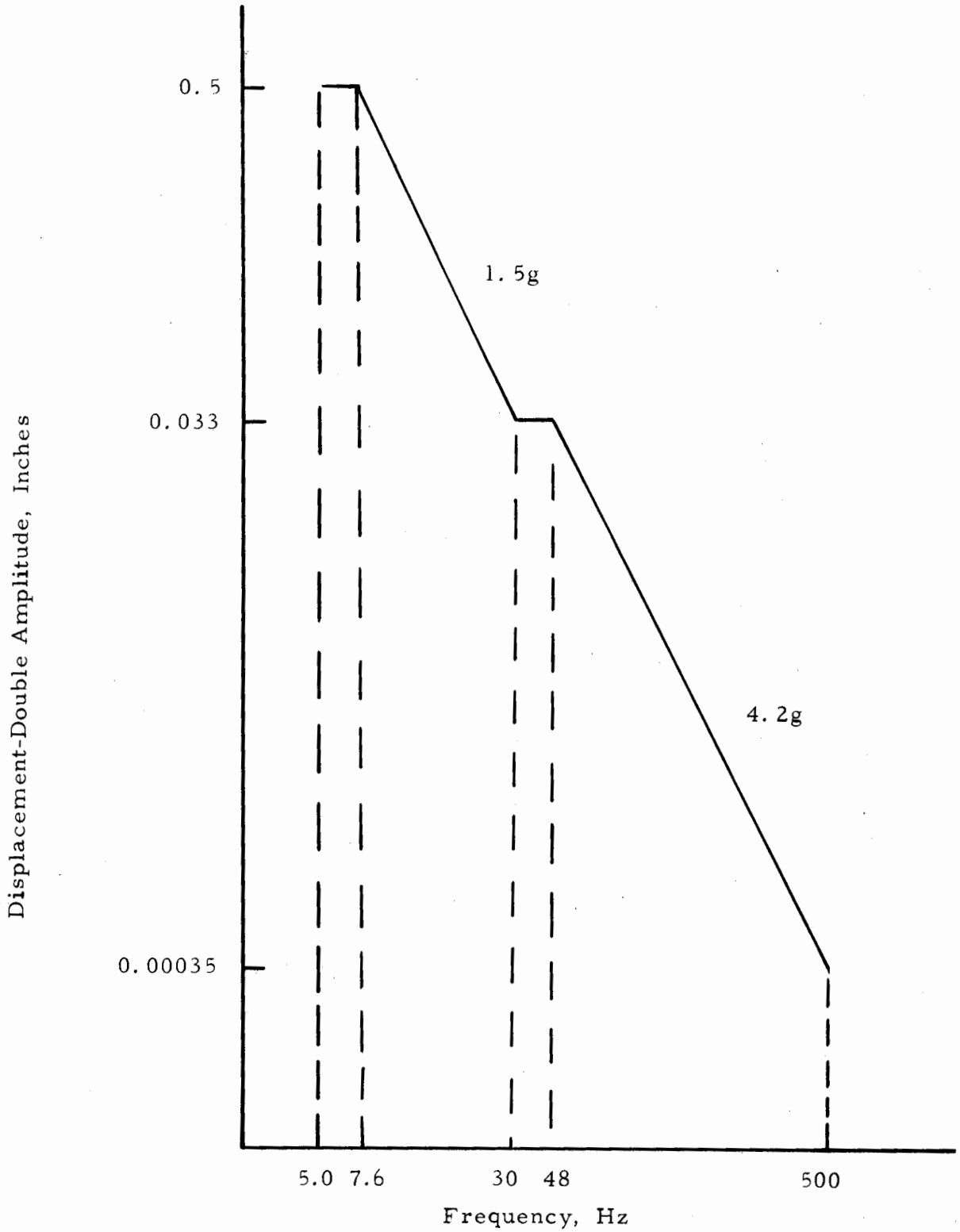


Figure 1. Vibration Envelope

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3.3.2.4 Vibration "B". The rangefinder shall show no physical failure and shall meet the requirements of 3.3.1.6b, 3.3.1.8, and 3.3.1.15 through 3.3.1.23 at ambient temperature after being exposed to the following vibratory conditions.

a.	Amplitude:	1/64 inch constant (1/32 inch double amplitude)
b.	Vibratory motion:	Simple harmonic
c.	Sweep time:	5 to 55 to 5 Hertz in a 5-minute period
d.	Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

3.3.2.5 Shock. The rangefinder shall meet the requirements of 3.3.1.6b, 3.3.1.8 and 3.3.1.15 through 3.3.1.23 after exposure to a total of 15 shock impulses, three in each direction along the three mutually perpendicular axis, except the one vertical plane (direction 12 to 6 o'clock with the receiver-transmitter in normal operating position). Each shock impulse shall be a half-sine wave with time duration of 3.5 ±0.5 msec and a peak amplitude of 100g.

3.3.2.6 Humidity. The rangefinder shall show no damage or deterioration which will impair performance after exposure to conditions of high relative humidity at an elevated temperature as specified in Method 507, Procedure I of MIL-STD-810 and shall meet the requirements of 3.3.1.3, and 3.3.1.8 while exposed to the conditions specified in Method 507, Procedure I, step 7 of MIL-STD-810.

3.3.2.7 Altitude. The rangefinder shall show no damage or deterioration which will impair performance after exposure to conditions of high altitude (low pressure) as specified in Method 500, Procedure I of MIL-STD-810 except that step 2 altitude limit shall be 10,000 feet and in step 3 change increase to maintain. The total exposure time for Procedure I shall be 5 hours. During exposure to the conditions of Step 3, the rangefinder shall meet the requirements of 3.3.1.3

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3.3.2.8 Dust. With the window cover closed, the rangefinder shall show no damage or deterioration after exposure to the dry dust (fine sand) laden atmosphere specified in Method 510 of MIL-STD-810. After exposure, the rangefinder shall meet the requirements of 3.3.1.3 and 3.3.1.8.

3.3.2.9 Salt fog. The rangefinder shall show no damage or deterioration after exposure to the salt fog atmosphere specified in Method 509 of MIL-STD-810. After exposure, the rangefinder shall meet the requirements of 3.3.1.3 and 3.3.1.8.

3.3.2.10 Fungus. The rangefinder shall show no damage or deterioration after exposure to an atmosphere containing a mixed spore suspension conducive to fungus growth as specified in Method 508 of MIL-STD-810. After exposure, the rangefinder shall meet the requirements of 3.3.1.3 and 3.3.1.8.

3.3.3 Electromagnetic compatibility. The rangefinder shall comply with the electromagnetic emission limits specified in MIL-STD-461 for Class IC equipment, except that the limits indicated by figures 2 through 8 herein shall apply.

3.3.4 Special operating conditions. The rangefinder, with the protective covers closed, shall withstand without damage, water hosing normally utilized to clean grime from combat vehicles.

3.3.5 Interchangeability. The rangefinder shall meet all the requirements of this specification without need for selective assembly of the units listed in table I. The units shall be interchangeable such that replacement of any single item or random combination of items of the rangefinder permits the rangefinder to operate and perform to the requirements of this specification.

3.3.6 Reliability. The rangefinder shall exhibit a mean ranging between failure (MRBF) of 5000 rangings at a 90 percent confidence level.

3.3.7 Sealing. With the rangefinder initially pressurized with nitrogen or air to 5.0 ± 1.0 pounds per square inch, gage (psig), the loss of pressure shall not exceed 0.22 psig in 2 hours.

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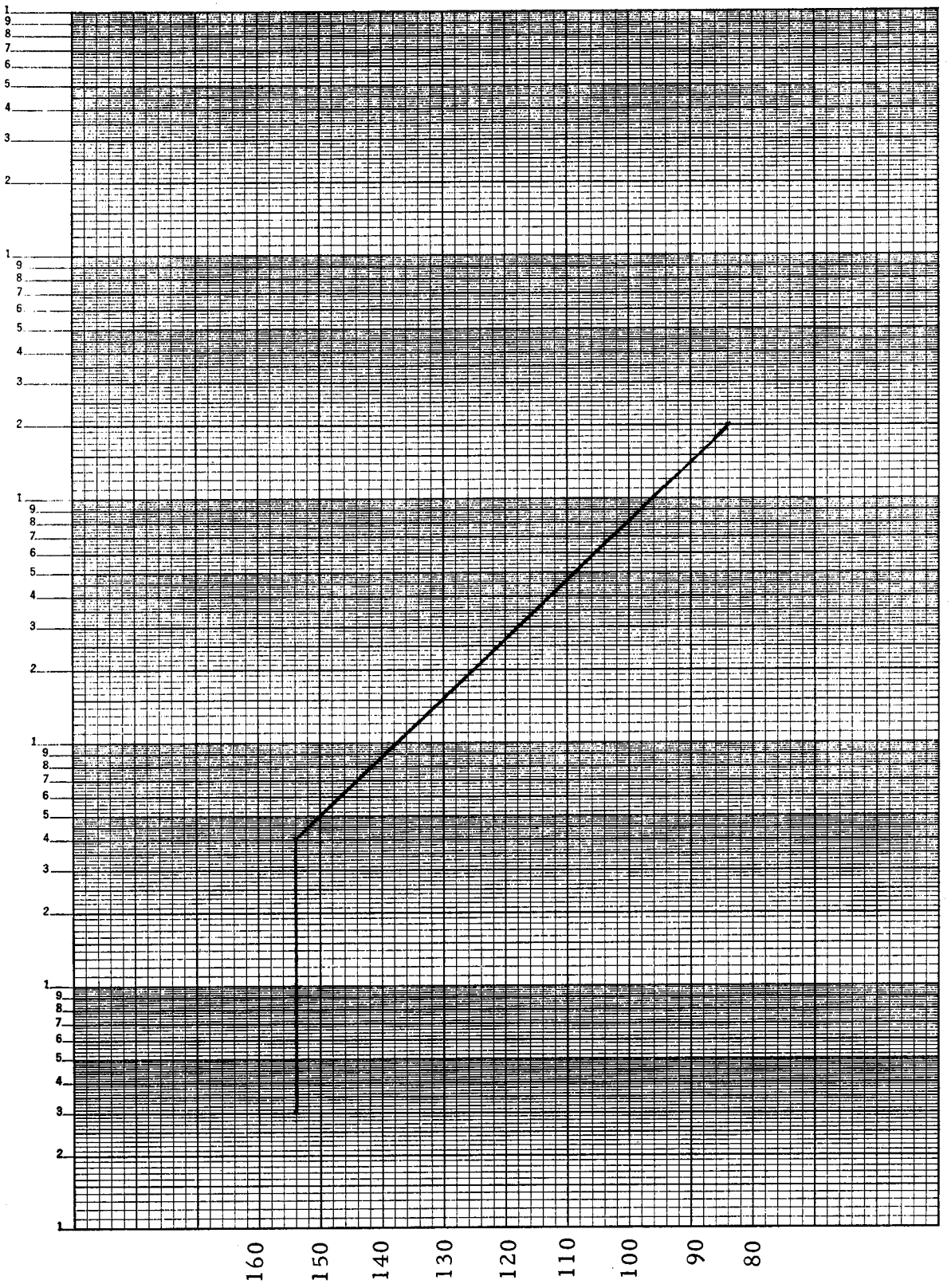


Figure 2. Limits for CE01 Narrowband Emissions

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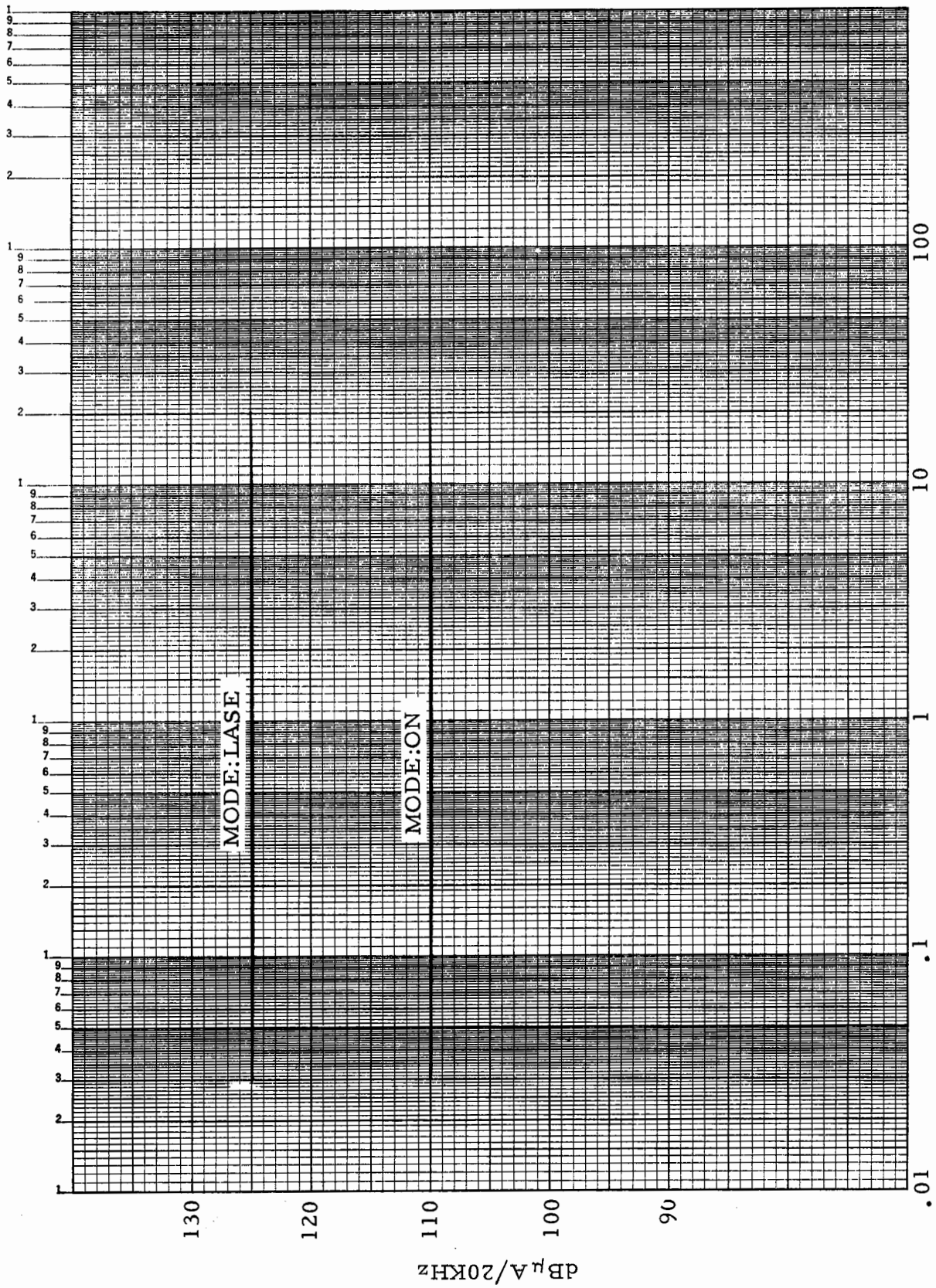


Figure 3. Limits for CEO1 Broadband Emissions

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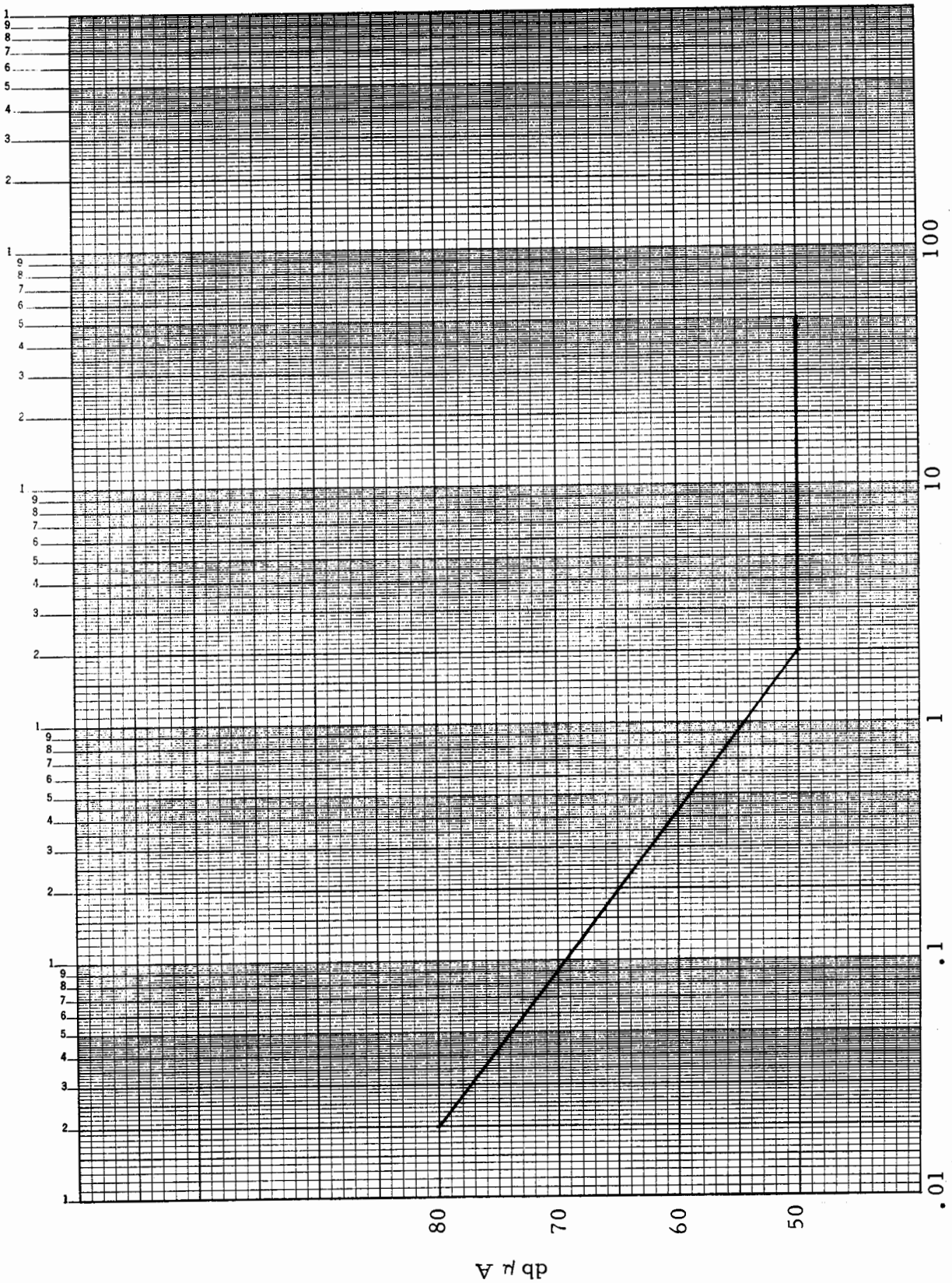


Figure 4. Limits for CE03 Narrowband Emissions

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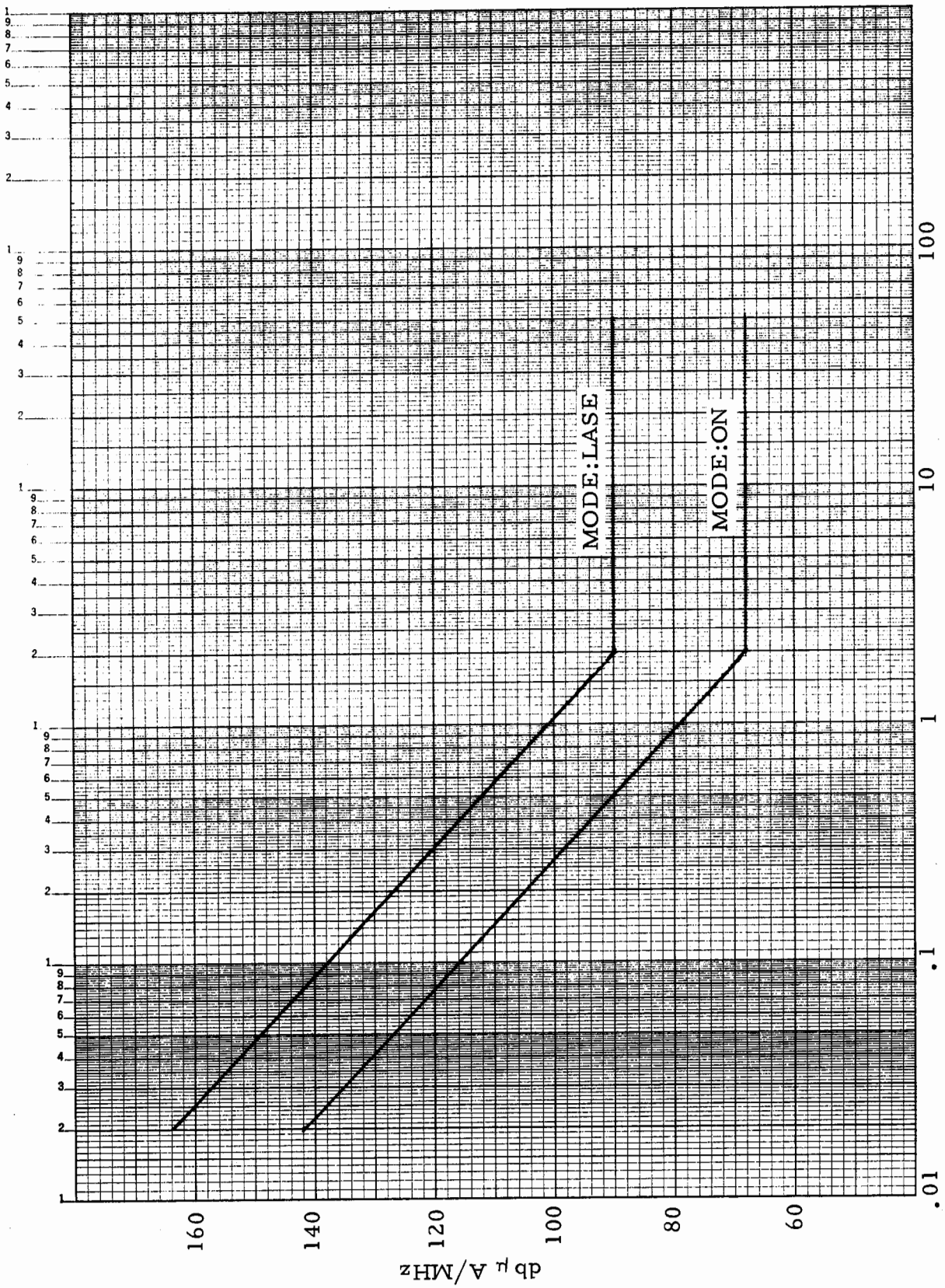


Figure 5. Limits for CEO3 Broadband Limits

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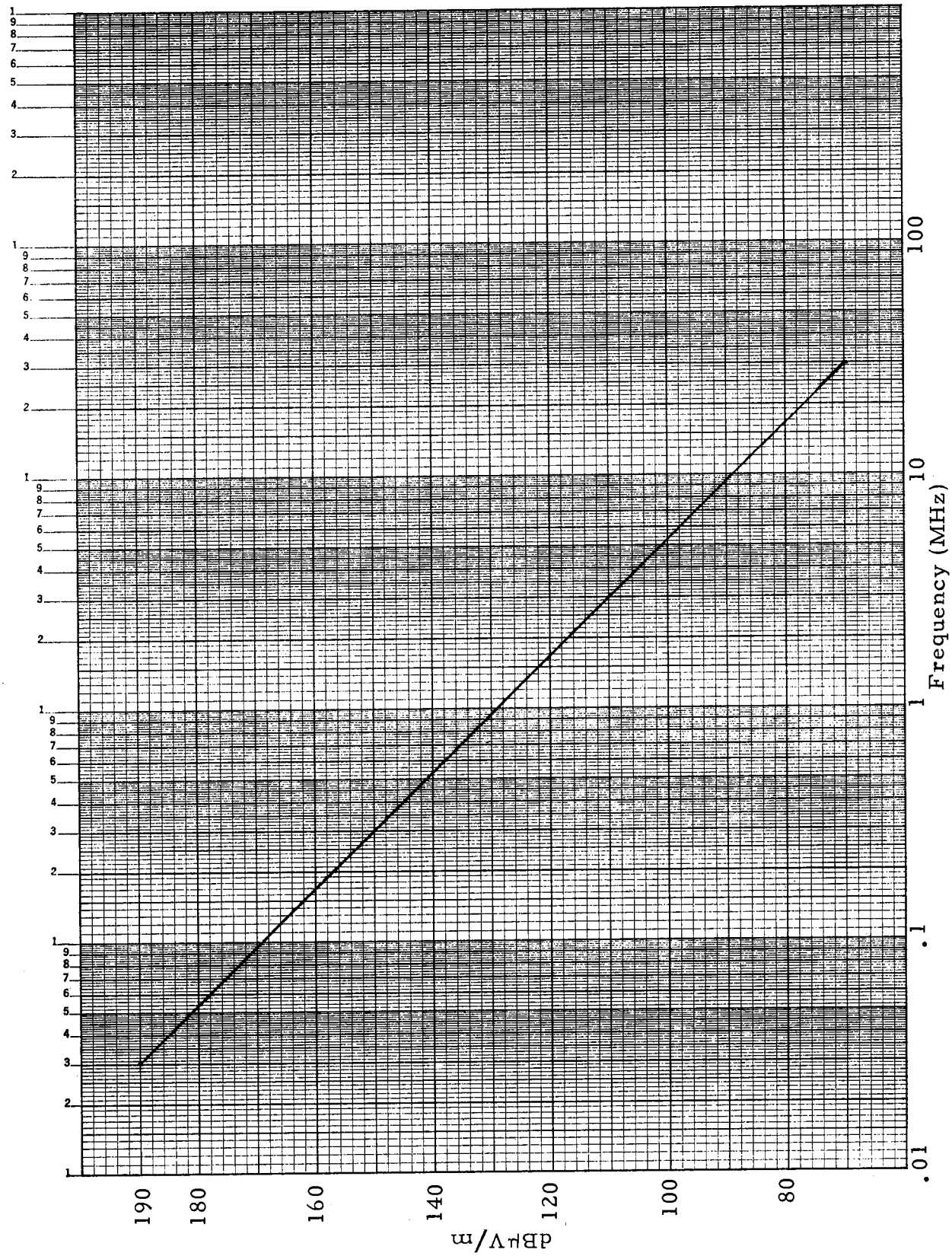


Figure 6. Limits for REO1

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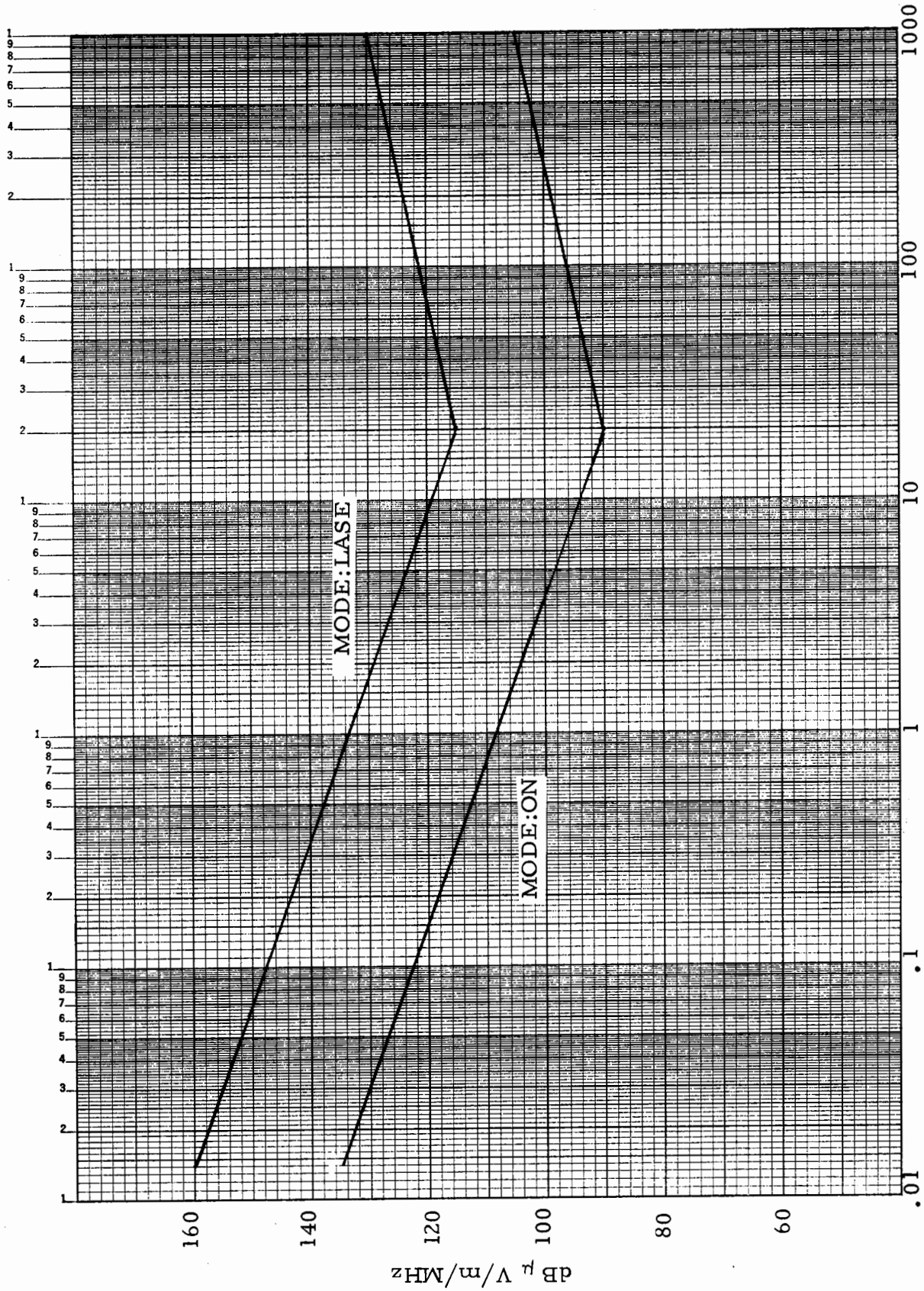


Figure 7. Limits for RE02 Broadband Emissions

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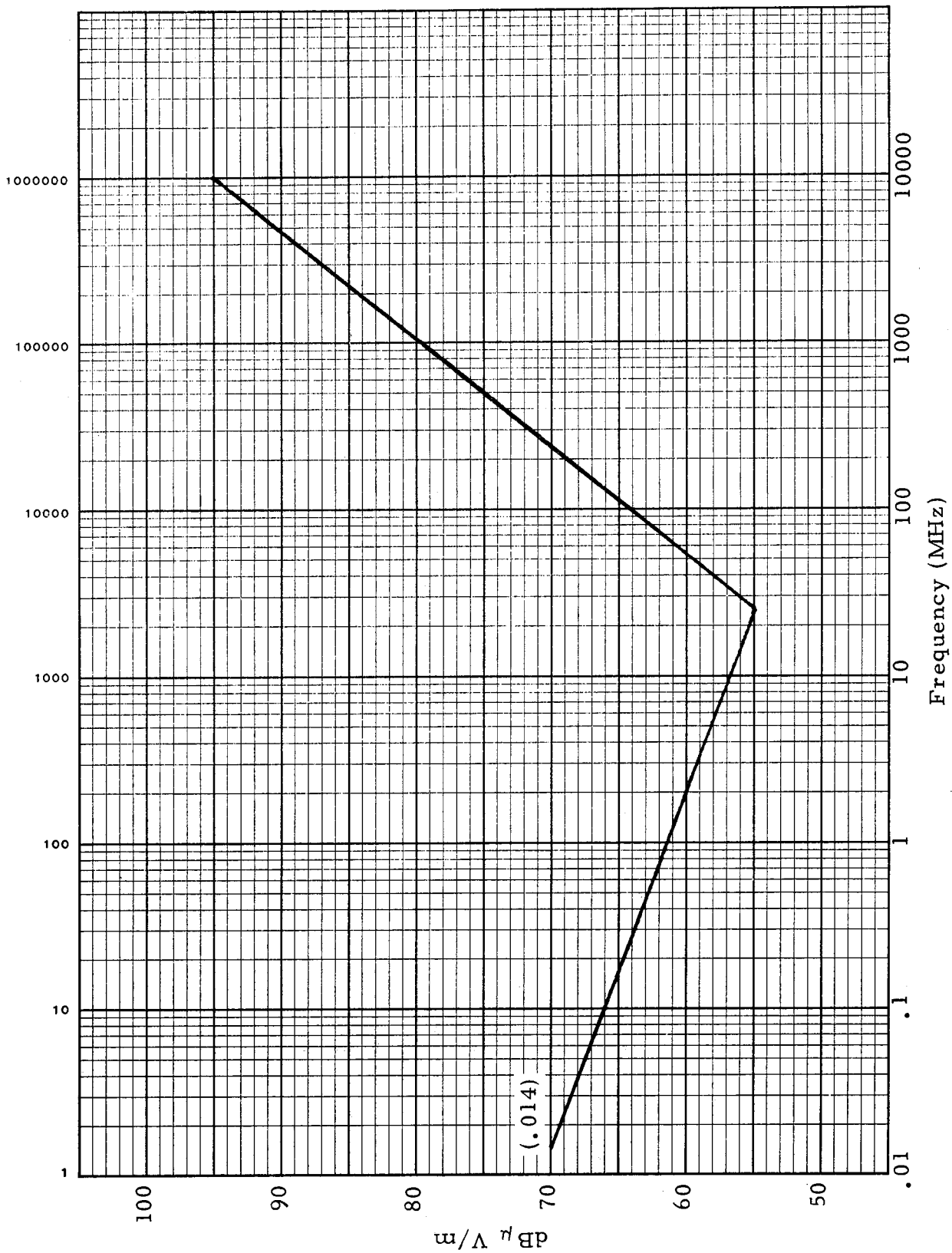


Figure 8. Limits for RE02 Narrowband Emissions

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3.3.8 Internal pressure. Subsequent to sealing, each unit shall be "dry" flushed* and an internal pressure of 3.5 ±0.5 psig shall be established. The receiver-transmitter unit shall be flushed with nitrogen in accordance with BB-N-411, Type I, Grade B, Class 1. All other units may be flushed with air.

*Dry flushing and pressurizing performed at the unit level may be utilized to fulfill this requirement.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 rangefinders plus 3 each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation,

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and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

4.3.2 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.3.3 Examination and tests

4.3.3.1 Components and subassemblies. All components and subassemblies listed in 3.1 shall be inspected in accordance with the inspection provisions contained in the respective specifications and SQAPs listed in the technical data package (TDP). In the absence of SQAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.3.2 First article testing. First article testing shall be conducted in accordance with tables III through VII, except vibration B, to determine compliance with 4.2. All examinations and tests shall be witnessed by the Government Quality Assurance Representative. When manufacturing processes, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability or reliability of the rangefinder, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

TABLE III
FIRST ARTICLE INSPECTION

Item	Characteristics	Requirements	Test Paragraph
1	Vibration "A"	3.3.2.3	4.6.25.2
2	Fungus	3.3.2.10	4.6.25.9
3	Electromagnetic Compatibility	3.3.3	4.6.26
4	Reliability	3.3.6	4.6.28

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4.3.3.3 First article selection. The Government Quality Assurance Representative shall randomly select from three groups of equipment, each group represented by the items listed in table I, for final assembly of three rangefinders. These three rangefinders shall be identified as rangefinders 1, 2, and 3 for first article testing.

NOTE: Subsequent to final assembly of each rangefinder, adjustment of PFN voltage shall be permitted, if necessary, to obtain the specified output.

4.3.3.4 Final acceptance. Final acceptance of the rangefinder and repair parts (when required by contract) shall be based upon compliance with all requirements (Section 3) and quality assurance provisions (Section 4) herein. Rejection shall be in accordance with 4.3.2 and as otherwise specified in the contract. Data obtained from lower level inspection tests may be applied in fulfilling the requirements of this specification.

4.3.3.5 Certification. Certification shall include test data and results to show evidence of compliance with 3.2. Certification shall be provided prior to performing first article testing and final acceptance inspection and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

4.3.3.6 Performance tests. The requirements and tests of table IV shall be inspected on a 100 percent basis. Unless otherwise specified, all examinations and tests shall be conducted at the standard ambient temperature to determine compliance with 3.3.1.

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One rangefinder shall be selected at random by a Government representative as a control sample from each 100 produced or from each months production, whichever occurs first. The control sample shall meet the requirements and tests in table V.

4.4.2 Follow-up tests. The rangefinders selected in 4.4.3 may be used to satisfy the requirements and tests of table VI at the specified intervals, after which the requirements and tests of table IV shall be met. These tests shall be conducted as follow-up tests to the first article samples and three each at 25 percent, 50 percent, and 75 percent completion of the production contract.

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TABLE IV
PERFORMANCE TESTS

Item	Characteristics	Requirement	Test Paragraph
101	Power turn-on condition	3.3.1.1	4.6.2
102	PFN charge and flashtube firing	3.3.1.2	4.6.3
103	Target selection	3.3.1.3	4.6.4
104	RESET switch	3.3.1.4	4.6.5
105	LASER MODE CONTROL switch TEST position	3.3.1.5.1	4.6.6.1
106	LASER MODE CONTROL switch "ON" position	3.3.1.5.2	4.6.6.2
107	Laser firing - Minimum range	3.3.1.6a	4.6.7.1
108	Laser firing - Maximum range	3.3.1.6b	4.6.7.2
109	Laser firing - Range accuracy	3.3.1.6c	4.6.7.3
110	Laser firing - Range resolution	3.3.1.6d	4.6.7.4
111	Laser firing - Minimum time between ranging	3.3.1.6e	4.6.7.5
112	Laser firing - Number of rangings (internal power)	3.3.1.6f(1) 3.3.1.6f(2)	4.6.7.6a 4.6.7.6b
113	Laser firing - Maximum number rangings	3.3.1.6g	4.6.7.7
114	Remote ranging	3.3.1.7	4.6.8
115	Peak output power	3.3.1.8	4.6.9
116	Input voltage variation	3.3.1.9	4.6.10
117	Maximum power drain	3.3.1.10	4.6.11
118	Primary power transient	3.3.1.11	4.6.12
119	Over voltage (malfunction)	3.3.1.12.1	4.6.13.1
120	"A" trigger reply (malfunction)	3.3.1.12.2	4.6.13.2
121	"A" trigger and counter (malfunction)	3.3.1.12.3	4.6.13.3
122	Battery charging (malfunction)	3.3.1.12.4	4.6.13.4
123	Dimming (DMR) switch	3.3.1.13	4.6.14
124	Test (TSW) switch	3.3.1.14	4.6.15
125	Sight / transmitter alignment	3.3.1.15	4.6.16
126	Optical scan angles	3.3.1.16	4.6.17

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TABLE IV (Continued)

Item	Characteristic	Requirement	Test Paragraph
127	Image tilt	3.3.1.17	4.6.18
128	Image wander	3.3.1.18	4.6.19
129	Reticle illumination	3.3.1.19	4.6.20
130	Battery drain indicator	3.3.1.20	4.6.21
131	Battery low indicator	3.3.1.21	4.6.22
132	Drive train torque and backlash	3.3.1.22	4.6.23
133	PFN discharge time	3.3.1.23	4.6.24

TABLE V
SPECIAL SAMPLING - GENERAL

Item	Control Test	Requirement	Test Paragraph
301.	Fabrication	3.1	Applicable drawings-visual
302.	General specification	3.2	MIL-F-13926 - visual

TABLE VI
FOLLOW-UP TESTS

Item	Characteristics	Requirement	Test Paragraph
303.	Shock	3.3.2.5	4.6.25.4
304.	Humidity	3.3.2.6	4.6.25.5
305.	Altitude	3.3.2.7	4.6.25.6
306.	Dust	3.3.2.8	4.6.25.7
307.	Salt fog	3.3.2.9	4.6.25.8
308.	Special operating conditions	3.3.4	4.6.27

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4.4.3 Environmental. Three rangefinders shall be selected at random from each 50 produced or each month's production, whichever occurs first. Each sample shall meet the requirements and tests in table VII and shall then meet all requirements and tests in table IV.

TABLE VII
SPECIAL SAMPLING - ENVIRONMENTAL

	Control Test	Requirement	Test Paragraph
309.	Storage temperature	3.3.2.1	4.6.25.1
310.	Operating temperature	3.3.2.2	4.6.25.1
311.	Vibration "B"	3.3.2.4	4.6.25.3
312.	Interchangeability	3.3.5	4.6.1.3
313.	Sealing*	3.3.7	4.6.29
314.	Internal pressure*	3.3.8	4.6.30
*To be performed after all environmental testing.			

4.4.4 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.2).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table VIII shall be utilized in performing the tests specified herein.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment

4.5.2.1 Government design. Unless otherwise specified in the contract all inspection equipment specified by drawing number in specifications or SQAPs forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the TDP.

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TABLE VIII
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	Oscilloscope with camera	Frequency response-dc to 150 megaHertz; input impedance-1 megohm or greater; sensitivity-5 mV/cm; sweep time-0.05 microseconds per centimeter; camera lens-f/1.3 (Tektronix C40 or equivalent); film-ASA equivalent of 10,000 (Polaroid Type 410 or equivalent).
2.	Dc power supply	0 to 90 Vdc, 15 A (24 Vdc nominal) Ripple: $\leq$200 mV p-p
3.	Dc voltmeter	Range: 0 to 100 Volts Accuracy: ±0.03 percent Input impedance: $\geq$1 megohm
4.	Dc ammeter	Range: 0 to 15 A Accuracy: ±5 percent
5.	Dc microammeter	Range: 0 to 100 μA Accuracy: ±1 percent Resistance: $\leq$ 300Ω
6.	Thermopile*	TRG 108, Tekmar Instruments, Inc. or equivalent
7.	Energy meter*	TRG 113, Tekmar Instruments, Inc. or equivalent
8.	Stop watch	0.5 second dial divisions; 60 seconds per revolution of long hand; ±10 percent accuracy
9.	Signal source	Capable of providing dc level as specified in table II, item 2.1
10.	Target No. 1	12-inch minimum diameter circle, 15 percent minimum diffuse reflectivity and located at 215 ±5 meters from window or receiver-transmitter. The exact location of the target within this tolerance shall be known within ±1 meter.

*An equivalent for items 6 and/or 7 is Model 580 Radiometer System EG&G.

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TABLE VIII (Continued)

Item	Nomenclature	Description
11.	Target No. 2	16-inch minimum diameter circle, 15 percent minimum diffuse reflectivity and located at the actual distance of Target No. 3 minus 20 meters, from window of receiver-transmitter. The exact location of the target within this tolerance shall be known within ±1 meter.
12.	Target No. 3	4 feet square minimum, 15 percent minimum diffuse reflectivity unless otherwise specified and located at 490 ±15 meters from window of receiver-transmitter. The exact location of the target within this tolerance shall be known within ±1 meter.
13.	Telescope	Three power (3X) minimum magnification with the crossline reticle pattern.
14.	Optical test target	Containing horizontal zero reference line (indicating vertical reference); a zero azimuthal reference line; ±2 degrees (from vertical) subtense lines (image tilt limits); a pair of near vertical lines, the separation of which subtends 2 degrees in azimuth at the rangefinder (image wander limit); ±10 degrees azimuth (vertical) lines; and -10 degree +20 degree elevation (horizontal) lines.
15	Lens, long focal length	Optical specialties 105-0829, or equivalent.
16	Travelling microscope	Unitron TMD (1" x 1"), or equivalent.

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4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

4.5.2.2.1 Pressure tester. Special test equipment (STE) in accordance with Drawing F8565556, or equivalent, shall be provided for testing the sealing of the rangefinder.

4.6 Test methods and procedures

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.1.1 Safety precautions. The following limitations are recommended for the protection of personnel against potential hazards from backscattered laser radiation.

a. No tests shall be conducted in rain, snow or hail.

b. Where possible, testing at night or day twilight shall be avoided; however, if necessary, all personnel using the rangefinder, or in the vicinity, shall be equipped with a laser safety glass of at least optical density 5, at 6943 angstroms.

c. Where possible, testing in daytime in heavy fog (visibility less than 200 meters) shall be avoided; otherwise, safety glasses as specified above shall be employed.

4.6.1.2 Initial conditions. Connect the power indicated in table II and, unless otherwise specified, allow the equipment to stabilize at an ambient temperature.

4.6.1.3 Interchangeability. The interchangeability requirements specified in 3.3.5 shall be satisfied when the rangefinder is selected as specified by 4.3.3.3 and passes the performance tests of table IV.

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NOTE: Whenever a malfunction is first indicated when conducting tests 4.6.1.4 through 4.6.28, press the manual RESET button and repeat the test to assure that a valid malfunction exists.

4.6.1.4 PFN voltage adjustment. Prior to operation, initially adjust R10 of the PFN charge power supply (10559345) for the PFN voltage determined by test and recorded for the transmitter component assembly (10559479), in accordance with MIL-C-14880. Readjust the PFN voltage for a laser energy output of 50 ±5 millijoules. The PFN current shall be 80 microamperes maximum.

4.6.2 Power turn-on condition test. Set the LASER MODE CONTROL switch to the ON position and observe the lighting of the pushbutton switches, the indicator readout, and the lighting of the LASER MODE CONTROL switch to determine compliance with 3.3.1.1.

4.6.3 PFN charge and flashtube firing test. Perform the following to determine compliance with 3.3.1.2:

a. When the RANGE light is flashing (system armed), press the RANGE button and listen for an audible sound caused by discharging of the PFN through the flashtube.

b. Press the RANGE button again and observe the RANGE button lighting sequence and the absence of sound from the flashtube.

c. Press the RESET button and observe lighting of the RANGE and LAST buttons.

4.6.4 Target selection test. Conduct the tests in steps 1 through 18 of table IX. The displays and readouts on the commanders control shall be as specified in table IX to insure compliance with 3.3.1.3.

4.6.5 RESET switch test. Conduct the tests in steps 1 through 8 of table IX to insure compliance with 3.3.1.4. (This test may be performed concurrently with the test of 4.6.4.)

4.6.6 LASER MODE CONTROL switch test

4.6.6.1 TEST position test. While conducting the tests in steps 1 and 2 of table IX,, listen for any audible sound from the flashtube to determine compliance with 3.3.1.5.1. The lighting of the RANGE button and target selection (3.3.1.3) shall also be observed to determine compliance with 3.3.1.5.1.

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TABLE IX

TARGET SELECTION TEST

Inputs		Indications		
Step Number	Press and Release Switches as Indicated	RANGE RETURN SELECTOR Lights	Range Returns	Range Readout (±15 Meters)
	Set LASER MODE CONTROL switch to TEST position			
1		LAST	0	0000
2	RANGE	LAST	1	845
3	RANGE	LAST	2	1845
4	RANGE	LAST	3	2845
5	RANGE	LAST	4	2845
6	"1"	"1"	4	845
7	"2"	"2"	4	1845
8	RESET	LAST	0	0000
	End of first test sequence.			
9	"1"	"1"	0	0000
10	RANGE	"1"	1	845
11	RANGE	"1"	2	845
12	RANGE	"1"	3	845
13	RESET	LAST	0	0000
	End of second test sequence.			
14	"2"	"2"	0	0000
15	RANGE	"2"	1	9995*
16	RANGE	"2"	2	1845
17	RANGE	"2"	3	1845
18	RESET	LAST	0	0000
	End of third test sequence.			

*If readout is 000 or 9990, reset system and repeat test sequence.

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4.6.6.2 ON position test. Conduct the tests in steps 1 through 3 of table X. The readouts on the commanders control shall be as specified in table X to determine compliance with 3.3.1.5.2.

4.6.7 Ranging - laser firing

4.6.7.1 Minimum range test. Perform five ranging operations against Target No. 1 (item 10 in table VIII) and observe the range readouts. The readouts shall be the actual target range within the specified range accuracy to insure compliance with 3,3.1.6a.

4.6.7.2 Maximum range test. Connect A77J5-r to A77J5-e and place neutral density filters over the receiver aperture with an attenuation of 37 ±1 dB. Perform 10 ranging operations against Target No. 3 (item 12 in table VIII). The target shall be flat, normal to the laser beam within ±10 degrees, with a minimum diffuse reflectivity of 80 percent. Observe the range readouts. At least 9 of the 10 readouts shall be the actual target range within the specified range accuracy to insure compliance with 3.3.1.6b.

4.6.7.3 Range accuracy test. Perform 5 ranging operations against Target No. 3 (item 12 in table VIII). Observe the accuracy of the indicated range to determine compliance with 3.3.1.6c.

4.6.7.4 Range resolution test. Perform 5 ranging operations against Targets No. 2 and 3 (items 11 and 12 in table VIII). The targets shall be aligned so as to allow the transmitted laser beam to strike both targets. The observed range readouts shall be the actual ranges to each of the targets within the specified range accuracy to insure compliance with 3.3.1.6d.

4.6.7.5 Minimum time between ranging test. Perform a ranging operation and measure the time the RANGE button light turns off and then turns on steady to determine compliance with 3.3.1.6e.

NOTE: In 4.6.7.6 and 4.6.7.7 there will be a MALF 1 readout and the MALF indicator will illuminate prior to each ranging operation.

4.6.7.6 Number of rangings using internal power source test.

a. Perform 30 ranging operations within a period of 10 minutes at a rate of 3 per minute against Target No. 3 (item 12 in

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TABLE X

LASER MODE CONTROL ON TEST

Inputs		Indications		
Step Number	Press and Release Switches as Indicated	RANGE RETURN SELECTOR Lights	Range Returns	Range Readout
Select a target beyond minimum range and within maximum range which is large enough to intercept the total laser beam with nothing visible between rangefinder and target.				
Set LASER MODE CONTROL switch to ON position.				
1		LAST	0	0000
2	RANGE	LAST	1	Range to target.
3	RESET	LAST	0	0000

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table VIII). Observe the indicated range accuracy of 3.3.1.6c to determine compliance with 3.3.1.6f (1).

b. Perform 10 ranging operations within a period of 4 minutes at a rate of 3 per minute against Target No. 3 (item 12 in table VIII) at -20°F with the battery fully charged. Observe the indicated range accuracy of 3.3.1.6c to determine compliance with 3.3.1.6f (2).

4.6.7.7 Maximum number of rangings test. Perform 12 ranging operations within a 2-minute period against Target No. 3 (item 12 in table VIII). (This may be performed concurrently with the test of 4.6.7.2). Observe the indicated range accuracy of 3.3.1.6c to determine compliance with 3.3.1.6g.

4.6.8 Remote ranging test. Set the LASER MODE CONTROL switch to the ON position and apply the signal source as specified in table II. Perform the test specified in table X, except do not press RANGE button. Observe the range readout to determine compliance with 3.3.1.7.

4.6.9 Peak output power test. The peak output power of the receiver-transmitter shall be computed for at least 2 firing operations to determine compliance with 3.3.1.8.

4.6.10 Input voltage variation test. Set the LASER MODE CONTROL switch to the ON position and perform five ranging operations against Target No. 3 at each of the following prime voltages: +24.0 ±2.0 Vdc, $+18^{+1}_{-0}$ Vdc, and $+30^{+0}_{-1}$ Vdc. Observe the accuracy of the range readouts at each input voltage to determine compliance with 3.3.1.9.

4.6.11 Maximum power drain test. Measure the prime input current to the rangefinder while performing steps 1 and 2 of table X, using the dc ammeter (item 4 in table VIII). Verify compliance with 3.3.1.10.

4.6.12 Primary power transient test. Perform steps 1 and 2 of table X to obtain a range readout on the commanders control. Reduce the primary input voltage to a value of $+6^{+1}_{-0}$ volts for a duration of 100 ±10 milliseconds and observe the commanders control readout to determine compliance with 3.3.1.11a. Apply a primary input voltage transient of 90^{+0}_{-5} volts for a duration of 100 milliseconds. Perform steps 1 and 2 of table X, then reapply the +90 volt, 100 ±10 millisecond

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transient and observe the commanders control readout to determine compliance with 3.3.1.11b.

4.6.13 Malfunction indication test. Except as otherwise noted, set the LASER MODE CONTROL switch to the TEST position and perform the following.

4.6.13.1 Overvoltage test. Set the LASER MODE CONTROL switch to the ON position and connect a 66.5k ±1.0 percent resistor between pins t and j on the test access connector A77J5. Observe the RANGE readout and MALF lamp to determine compliance with 3.3.1.12.1.

4.6.13.2 "A" trigger reply test. Place neutral density filters over the receiver aperture with an attenuation greater than 80 dB and perform steps 1 and 2 of table X. Observe the MALF lamp and RANGE readout to determine compliance with 3.3.1.12.2.

4.6.13.3 "A" trigger and counter test. Depress the RANGE button for at least 1 second and observe the MALF lamp and RANGE readout to determine compliance with 3.3.1.12.3.

4.6.13.4 Battery charging test. Interrupt the input voltage (item 1.1, table II) for a period less than 3 seconds and observe the MALF lamp and RANGE readout to determine compliance with 3.3.1.12.4.

4.6.14 DMR switch test. With the indicator lamps lighted as in 4.6.2, depress the DMR switch and observe indicator lamp response to determine compliance with 3.3.1.13.

4.6.15 TSW switch test. Manually depress the TSW switch on the commanders control and observe the RANGE and RETURNS readouts and the indicator lamps to determine compliance with 3.3.1.14.

4.6.16 Sight/transmitter beam alignment test. Mount the lens (item 15, table VIII) in front of the r-t unit window. Locate in the focal plane of the lens a piece of thermally sensitive paper containing target grids. Sight through the eyepiece and position the receiver reticle symetrically over the juncture of a target grid. Fire the laser. Position the receiver reticle over the juncture of four more target grids and fire the laser at each of these target grids in turn for a total of five firings. Dismount the thermally sensitive paper. Measure with a microscope (item 16, table VIII) the average deviation of the centers of the five burn spots with respect to the target grid junctures to determine compliance with 3.3.1.15.

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NOTE: The tests in 4.6.17 through 4.6.19 may be performed at the unit level or only with receiver-transmitter and receiver-transmitter control.

4.6.17 Optical scan angles test. Rotate the azimuth and elevation controls on the receiver- transmitter control to the zero position. Place a test pattern (item 14, table VIII) with reference scan angle designations for azimuth (±10°) and elevation (-10°, +20°) in front of the rangefinder. Rotate the azimuth control in maximum clockwise and counterclockwise positions and observe the center of the reticle pattern with respect to the established horizontal reference. Rotate the azimuth control to zero position and repeat the test for elevation control to determine compliance with 3.3.1.16.

4.6.18 Image tilt test. Perform this test utilizing an auxiliary viewing telescope (item 13, table VIII) and a calibrated target (item 14, table VIII). Position the target in front of the rangefinder telescope with the vertical reference lines plumb and the zero reference point aligned with the telescope line of sight (azimuth and elevation scan controls set at zero). Mount the auxiliary telescope with the vertical reticle line plumb at the rangefinder telescope eyepiece assembly, and center the intersect point on the telescope reticle on the target zero reference point. Rotate the elevation scan control over the limits (-10°, +20°) and view the target image through the auxiliary telescope to determine compliance with 3.3.1.17.

4.6.19 Image wander test. This test shall be conducted in conjunction with the test in 4.6.18. While observing the established position (intersecting point) of the auxiliary telescope reticle, rotate the elevation scan control of the rangefinder as specified in 3.3.1.18. Observe the image wander of the target as a result of elevation and depression to determine compliance with 3.3.1.18.

4.6.20 Reticle illumination test. Adjust the reticle light control knob to obtain normal reticle illumination. Visually examine the illuminated reticle etchings against a darkened background to determine compliance with 3.3.1.19. Rotate the reticle knob from one extreme to the other and observe the variation in intensity of reticle illumination to determine compliance with 3.3.1.19.

4.6.21 Battery drain indicator test. Set the LASER MODE CONTROL switch to the ON or TEST position. Reduce the input voltage to +16.5 ±0.5 volts. Observe the battery drain indicator to determine compliance with 3.3.1.20.

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4.6.22 Battery low indicator test. Set the LASER MODE CONTROL switch to the ON or TEST position. Connect A77J5-E to A77J5-e. Observe the battery low indicator to determine compliance with 3.3.1.21.

4.6.23 Drive train torque and backlash test.

NOTE: The tests in 4.6.23.1 and 4.6.23.2 may be performed at the unit level or only with the receiver-transmitter and receiver-transmitter control.

4.6.23.1 Azimuth test. Apply starting torque to the azimuth positioning knob of the receiver-transmitter control. Monitor the movement of the scan prism (optically or mechanically). Verify that the requirements of 3.3.1.22.2 are met.

4.6.23.2 Elevation test. Apply starting torque to the elevation positioning knob of the receiver-transmitter control. Monitor the movement of the scan prism (optically or mechanically). Verify that the requirements of 3.3.1.22.2 are met.

4.6.24 PFN discharge time test. Connect a microammeter between pins t(+) and e(-) on test access connector J5 on the Laser Power Supply Control (11738831). Remove the power source (item 1.1 of table II) by setting the LASER MODE CONTROL switch to the OFF position. Measure the PFN current decrease to determine compliance with 3.3.1.23.

4.6.25 Environmental tests.

NOTE: Environmental testing of the separate components, assemblies, and units, including spares, of the fire control rangefinder shall not be required when the rangefinder is tested as a fire control rangefinder subsystem as specified herein.

4.6.25.1 Temperature tests. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926, except that assemblies or units may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F) with an additional operating condition at -20°F, and with the rate of temperature change during cycling not greater than 8°F per minute. Perform the tests of 4.6.4, 4.6.7.3, and 4.6.9 at -40°F and +125°F, and the test of 4.6.7.6b at -20°F to determine compliance with 3.3.2.2. Perform the tests of 4.6.4, 4.6.7.2, 4.6.9, and 4.6.16 through 4.6.24 at ambient temperature after exposure to and thermal stabilization at -80°F and +160°F to determine compliance with 3.3.2.1.

4.6.25.2 Vibration "A" test. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Procedure VIII therein except the vibratory conditions of 3.3.2.3

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shall apply. During vibration exposure, perform the test of 4.6.4 to determine compliance with 3.3.2.3. Subsequent to vibration exposure, examine the rangefinder for structural failures, loose hardware, wire breakage, abrasion, and contamination of optical components. Perform the tests of 4.6.7.2, 4.6.9, and 4.6.16 through 4.6.24 to determine compliance with 3.3.2.3.

4.6.25.3 Vibration "B" test. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part I of Procedure I therein, except that the vibratory conditions of 3.3.2.4 shall apply. Following vibration exposure, examine the assembly for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.7.2, 4.6.9, and 4.6.16 through 4.6.24 to determine compliance with 3.3.2.4.

NOTE: The shock test specified in 4.6.25.4 may be performed at the unit level.

4.6.25.4 Shock test. Perform this test utilizing the apparatus specified in Method 516.1 of MIL-STD-810, in accordance with Procedure I therein, except the conditions of 3.3.2.5 herein shall apply. After the shocks, perform a visual-tactile examination for damage and loose parts. Perform the tests of 4.6.7.2, 4.6.9, and 4.6.16 through 4.6.25 to determine compliance with 3.3.2.5.

4.6.25.5 Humidity test. Perform this test utilizing apparatus specified in Method 507 of MIL-STD-810, in accordance with Procedure I therein and with the conditions specified in 3.3.2.6. During step 7 of Procedure I, perform the tests of 4.6.4, and 4.6.9 to determine compliance with 3.3.2.6.

4.6.25.6 Altitude test. Perform this test utilizing apparatus specified in Method 500 of MIL-STD-810, in accordance with Procedure I therein and with the conditions specified in 3.3.2.7. During step 3 of Procedure I, perform the tests of 4.6.4 to determine compliance with 3.3.2.7.

4.6.25.7 Dust test. Perform this test utilizing the apparatus specified in Method 510 of MIL-STD-810, in accordance with Procedure I therein, except step 5. After exposure, perform the test of 4.6.4 and 4.6.9 to determine compliance with 3.3.2.8.

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4.6.25.8 Salt fog test. Perform this test utilizing the apparatus (chamber, atomizers, air supply, and salt solution) specified in Method 509 of MIL-STD-810, in accordance with Procedure I therein, except the rangefinder shall not be operated during performance of test. After exposure, perform the tests of 4.6.4 and 4.6.9 to determine compliance with 3.3.2.9.

4.6.25.9 Fungus test. Perform this test utilizing the apparatus specified in Method 508 of MIL-STD-810, in accordance with Procedure I therein with the unit nonoperating. After exposure, perform the tests of 4.6.4 and 4.6.9 to determine compliance with 3.3.2.10.

4.6.2.6 Electromagnetic compatibility test. Electromagnetic measurements shall be performed in accordance with the applicable procedures specified in MIL-STD-462 for Class IC equipment to determine compliance with 3.3.3.

4.6.27 Special operating conditions test. Prior to performing this test, all units shall be protected from direct impingement of water blast by using protective devices simulating the ballistic protective covers used when installed in the M551 vehicle. Perform this test utilizing a commercial type high pressure water hose with a water pressure equivalent to that from a standard fire hydrant. The direction of force shall be from each of 3 mutually perpendicular planes oriented to the center of mass initially and slowly traversing the entire area. Each water application shall be for a period of at least 3 minutes with a distance of 5 feet $\pm$6 inches from nozzle to applied surface. A visual and tactile examination shall be conducted for loose or broken parts and other possible damage to determine compliance with 3.3.4.

4.6.28 Reliability test. Reliability testing shall be performed in conjuction with First Article Testing. During those rangefinder firing tests outlined in this specification, additional firings shall be performed in order that the rangefinder reliability (3.3.6) may be verified. Each sample rangefinder shall accumulate a minimum of 3000 rangings (First Article sample to three (3) rangefinders). A record shall be maintained for each sample indicating serial number, type of environmental exposure, number of failures (type and cause) (see 6.3.2), and range data (true range and rangefinder range), as well as type of power supply and battery used (i.e., rangings per battery, serial number before requiring each recharge and the number of rechargings required during tests). An entry will be

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made into the record each time the rangefinder is fired. The data collected will be reduced and compared to the following relationship:

$$5000 \leq \frac{2T}{\chi^2.10;\ 2(r+1)}$$

where

T	=	Total number of rangings
r	=	Number of chargeable reliability failures
$\chi^2.10;\ 2(r+1)$	=	Chi Square 90 percent lower value with 2(r+1) degrees of freedom.

Subsequent to the reliability test, the sample rangefinder shall be refurbished and readied for shipment and delivery.

4.6.28.1 Lasing termination. Reliability lasings shall be terminated when any one of the following conditions are met:

a. Upon successful completion of the tests defined in 4.6.28,

b. When 6 failures occur, or

c. When 46, 373 lasings are accumulated.

NOTE: Termination shall be based upon whichever of a, b, or c occurs first.

4.6.29 Sealing test. Perform this test utilizing the STE specified in 4.5.2.2.1. Pressurize the rangefinder as specified in 3.3.7. Measure the pressure loss over a minimum period of 2 hours to determine compliance with 3.3.7.

4.6.30 Internal pressure test. Perform this test utilizing the STE specified in 4.5.2.2.1. After dry flushing, pressure the receiver-transmitter. Verify that the requirements of 3.3.8 are met.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11738801. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. The Fire Control Rangefinder AN/VVG-1 ( ) is a component of the fire control system used in conjunction with the M551 Armored Reconnaissance/Airborne Assault Vehicle (Sheridan). The function of the rangefinder is to improve the first round hit capability of the conventional round.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Applicable packaging data sheet (see 5.1)

c. Selection of applicable levels of preservation, packaging and packing

d. Applicable stock number

e. Requirements for First Article submission which may include additional tests selected by the procuring agency.

6.3 Definitions.

6.3.1 Peak power. Peak power is defined as the ratio of the total laser output energy to the duration of the laser pulse at the one-half peak power points.

6.3.2 Failure. A failure is defined as any malfunction which renders the rangefinder inoperative or operative outside the limits of this specification. In the determination of "chargeable" reliability failures the following clarification will apply:

a. Upon failure analysis (to determine the cause of any malfunction), it is determined than an "assignable cause" (related to manufacturing procedures, basic design weakness, etc.) was the factor which caused the occurrence of the malfunction, the malfunction will not be chargeable

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as a reliability failure; however, the correction to the manufacturing process, or procedure, et al, shall be accomplished and the testing resumed at the environment at which the malfunction occurred.

b. Upon failure analysis no assignable cause is evident and the malfunction is considered random in nature; the malfunction shall be a chargeable reliability failure and, as such, shall be considered in the demonstration equation (4.6.28).

6.3.2.1 Rangefinder interruption. Rangefinder interruption is any rangefinder malfunction for which the rangefinder is operable within the limits set forth in this specification but requires maintenance to correct the malfunction. Although an interruption does not penalize the reliability, it may constitute a failure of some major subassembly for which rangefinder acceptance cannot be made until that characteristic or component which caused the interruption has been screened.

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-50749(MU) Rangefinder, Fire Control: AN/VVG-1() 11738801

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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